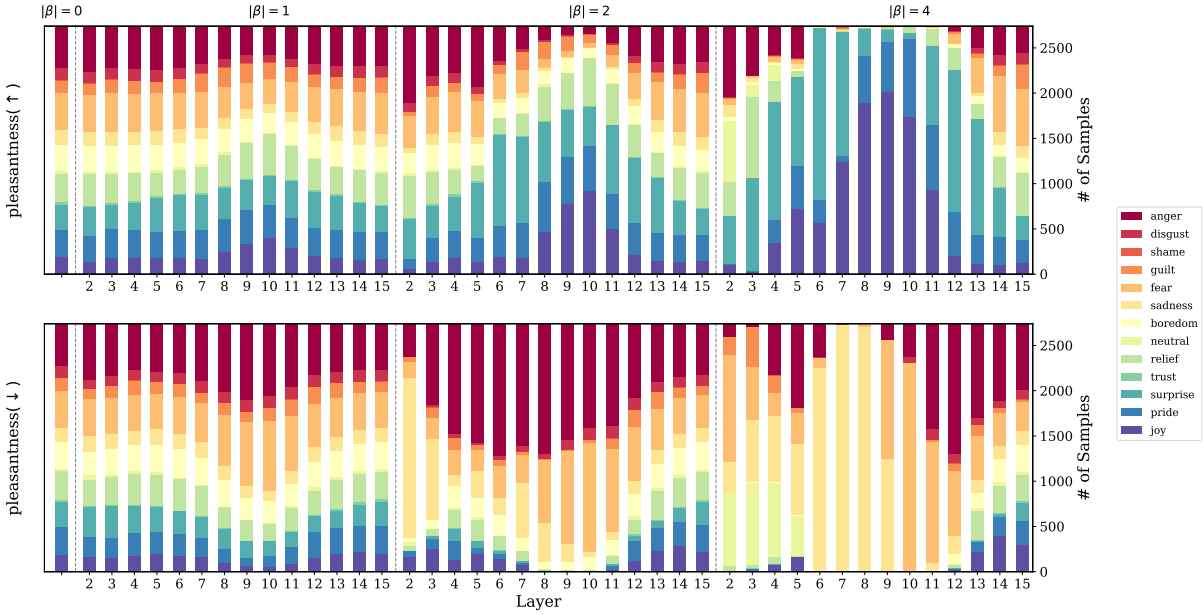

Figure 30: Effect of promoting and demoting *pleasantness* at different layers of Llama 3.2 1B with three levels of scaling factor β. $\beta = 0$ represents the original distribution without appraisal modulation. A consistent increase in distribution shift is observed as β increases across all intervention experiments. However, when intervening on earlier layers, particularly at higher β values, the shift in the distribution of represented emotions does not always align with theoretical expectations.

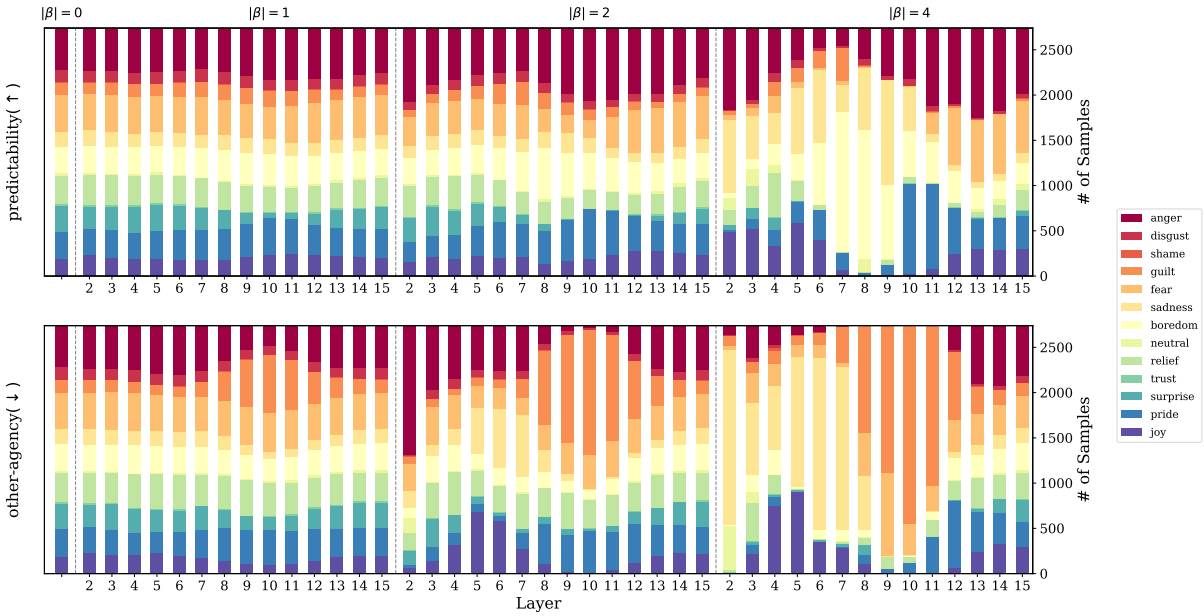

Figure 31: Effect of promoting and demoting *other-agency* at different layers of Llama 3.2 1B using three levels of scaling factor β. $\beta = 0$ represents the original distribution without appraisal modulation. Mid-layer appraisal modulation exhibits a theoretically plausible shift in emotion distribution.

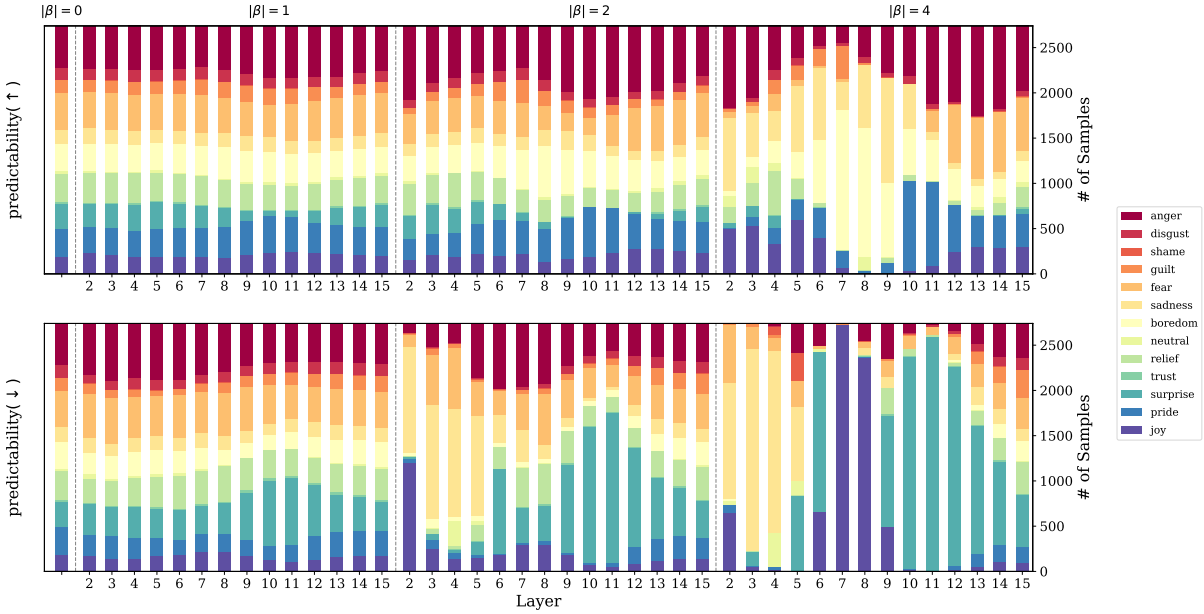

Figure 32: Effect of promoting and demoting *predictability* at different layers of Llama 3.2 1B using three levels of scaling factor β. $\beta = 0$ represents the original distribution without appraisal modulation. Mid-layer appraisal modulation exhibits a theoretically plausible shift in emotion distribution.

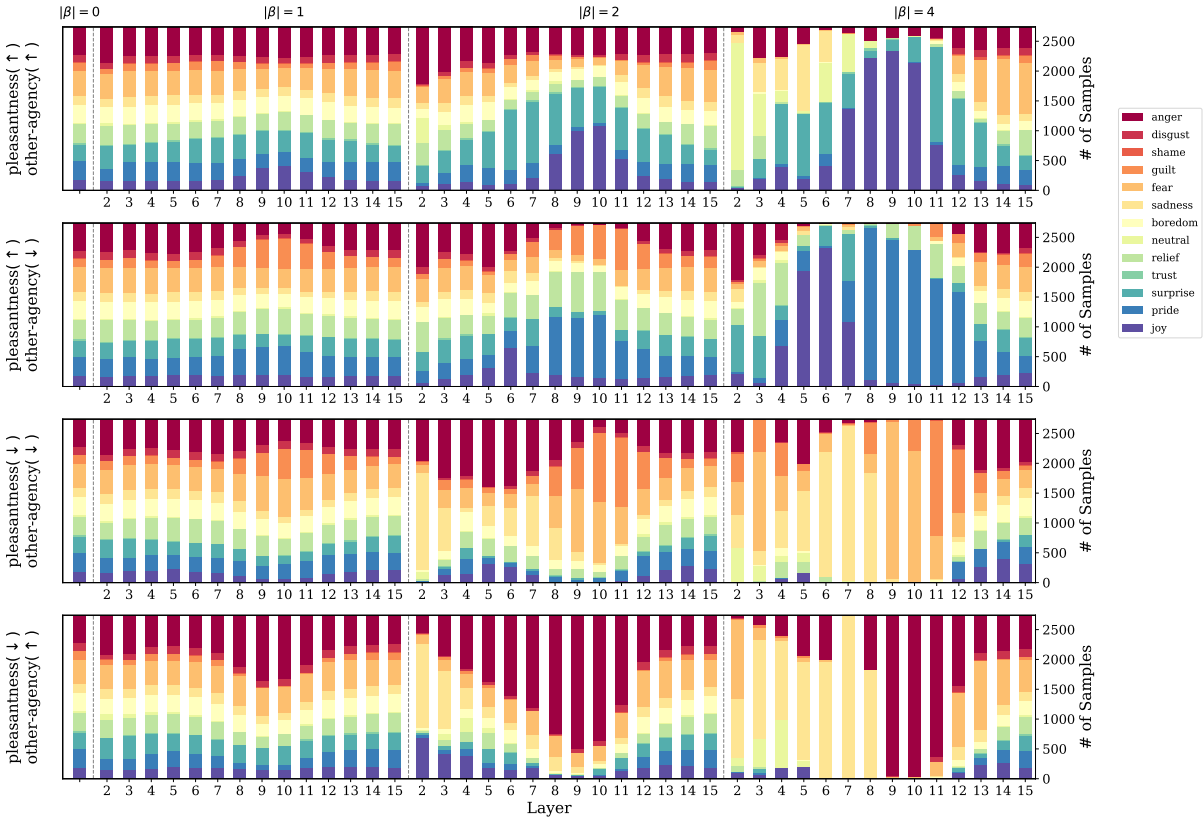

Figure 33: Superposition of *pleasantness* and *other-agency* appraisal modulation at different layers of Llama 3.2 1B. Results show successful promotion of *pride* with no further occurrences of *joy* in layers 9–11 when demoting *other-agency*, and the promotion of *guilt* and *fear* with no occurrences of *anger* in mid-layers when demoting *other-agency*.

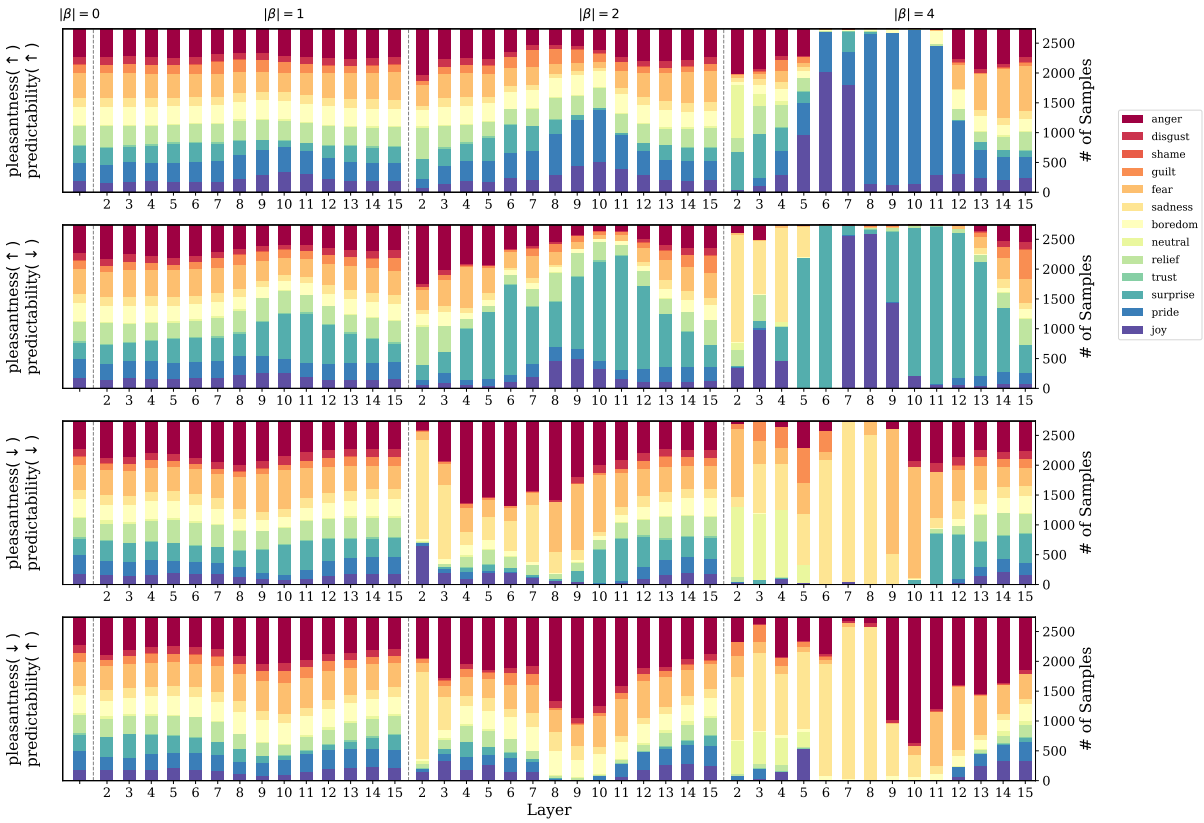

Figure 34: Superposition of *pleasantness* and *predictability* appraisal modulation at different layers of Llama 3.2 1B. Results show a successful transition from *pride* to surprise and a greater distribution of *fear* compared to anger in mid-layers when promoting unpredictability.

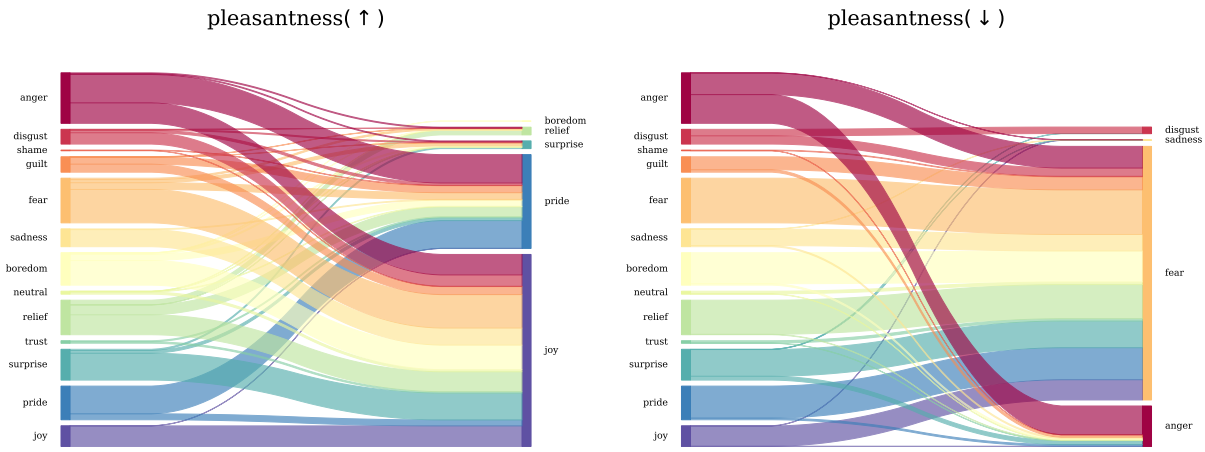

Figure 35: Sankey plot for *pleasantness* appraisal modulation when we perform it at layer 9 of Llama 3.2 1B model.